Soft-Prompt Conditioned ESM-2 for Multi-Task Antimicrobial Resistance Prediction

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Background and Context

Antimicrobial resistance (AMR) is a major global health threat, linked to an estimated 1.27 million deaths worldwide in 2019 [1]. Traditional AMR gene annotation relies on sequence alignment against reference databases (e.g., BLAST), which struggles with novel sequences - exactly those most likely to signal emerging resistance.

Protein language models (PLMs) like ESM-2 learn protein function directly from sequence, without needing a close database match. We use ESM-2 as a **frozen** backbone, paired with a lightweight module that steers its representations toward task-relevant context.

[1] Antimicrobial Resistance Collaborators. "Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis." *The Lancet*, vol. 399, no. 10325, 2022, pp. 629–655.

Objective

To test how soft-prompt conditioning shapes frozen ESM-2's AMR predictions across tasks of varying signal strength, evaluated via an internal-vs-external soft-prompt injection-mode ablation.

Significance

This work tests whether biological metadata can steer a frozen PLM's predictions without fine-tuning it, using **soft prompting** - a small set of learnable vectors injected into the model. This stays lightweight compared to full fine-tuning, while preserving ESM-2's existing sequence understanding.

Results

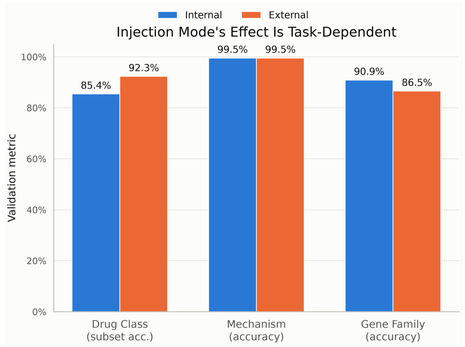


Figure 2:
Injection mode's effect is task-dependent: external injection improves drug-class prediction (+7pp), meanwhile internal injection preserves gene-family accuracy under weak conditioning, and neither moves mechanism.

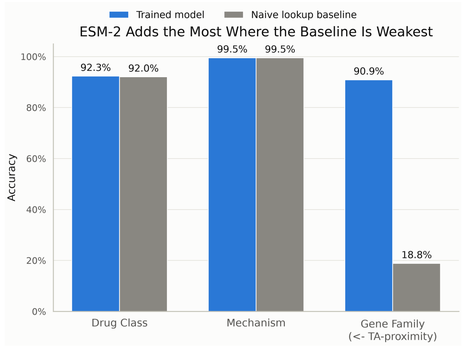


Figure 3:
ESM-2 barely outperforms a naive conditioning-only baseline on drug class and mechanism, both are nearly tied, but adds ~72 points on gene family, where that baseline is weakest.

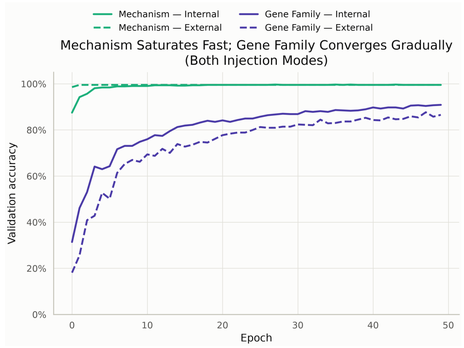


Figure 4:
Mechanism accuracy crosses 98% within the first few epochs and plateaus by ~epoch 18, while gene-family accuracy converges gradually across all 50.

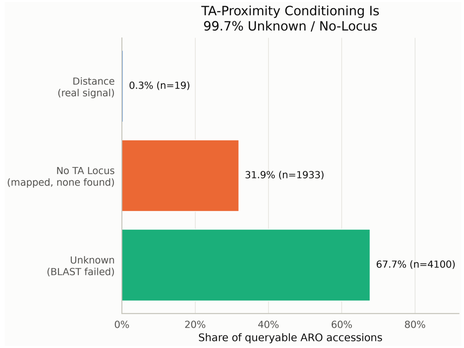


Figure 5:
TA-proximity conditioning is 99.7% unknown/no-locus, a real coverage gap between CARD and RefSeq/TADB 3.0, not a design choice, and the reason gene family's baseline is so weak.

Architecture

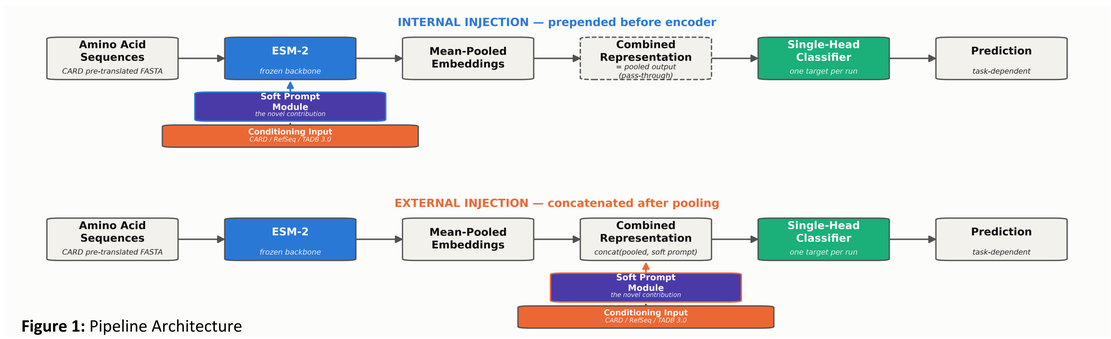


Figure 1: Pipeline Architecture

Label Leakage Confound

V1 conditioned two heads on their own ground-truth labels, letting them shortcut ESM-2 entirely. Accuracy hit near-ceiling by epoch 0 - fixed by ensuring conditioning input and prediction target are never the same field.

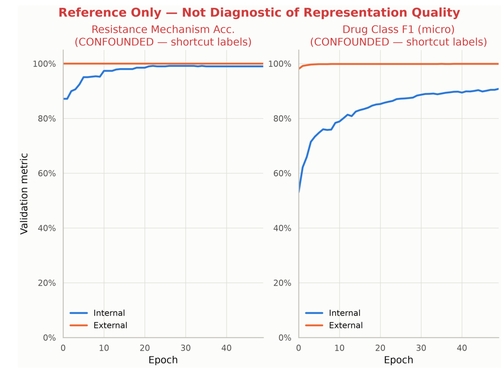


Figure 6: Label Leakage Confound

Conclusions

Injection mode has no universal winner, since the best-performing mode depends on the task rather than the architecture. ESM-2's entire measurable contribution over a naive baseline is concentrated in gene family classification, where it beat the baseline by roughly 72 points. That gap traces back to data coverage - only 0.31% of ARO accessions mapped to a real TA-locus distance - leaving gene family as the one task with no metadata shortcut to lean on.

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